MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, GENERAL LIGHTING,

45 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 82, 82.1, 343.1 AND 343.2

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

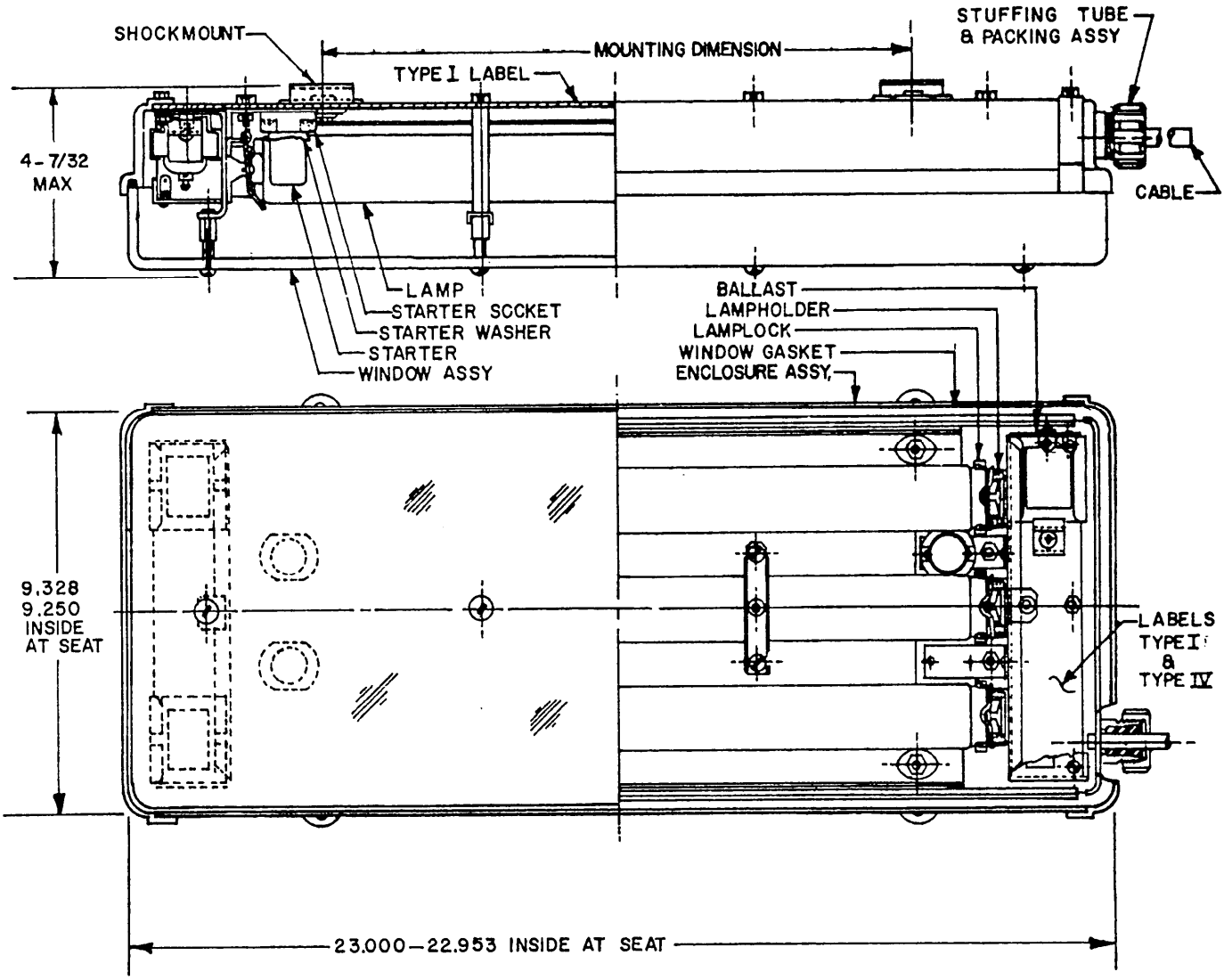


FIGURE 1. Typical view of Symbols 82 and 343.1.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6210

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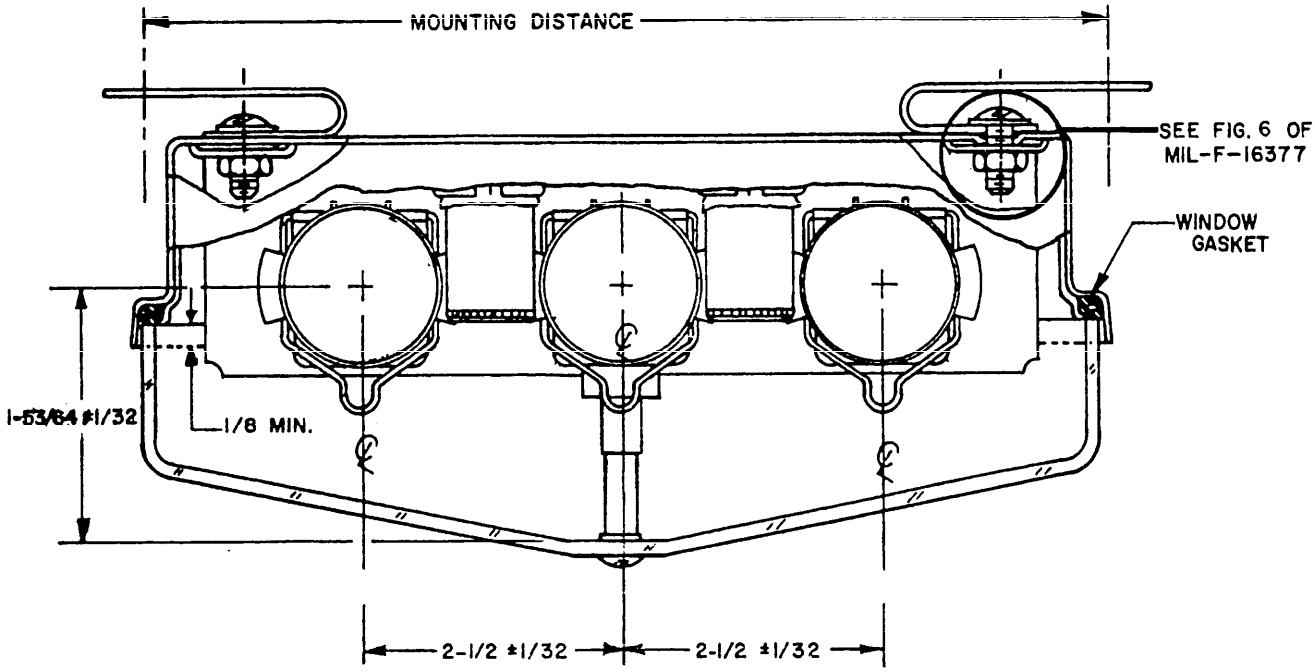


FIGURE 2. Typical section for Symbols 82 and 343.2.

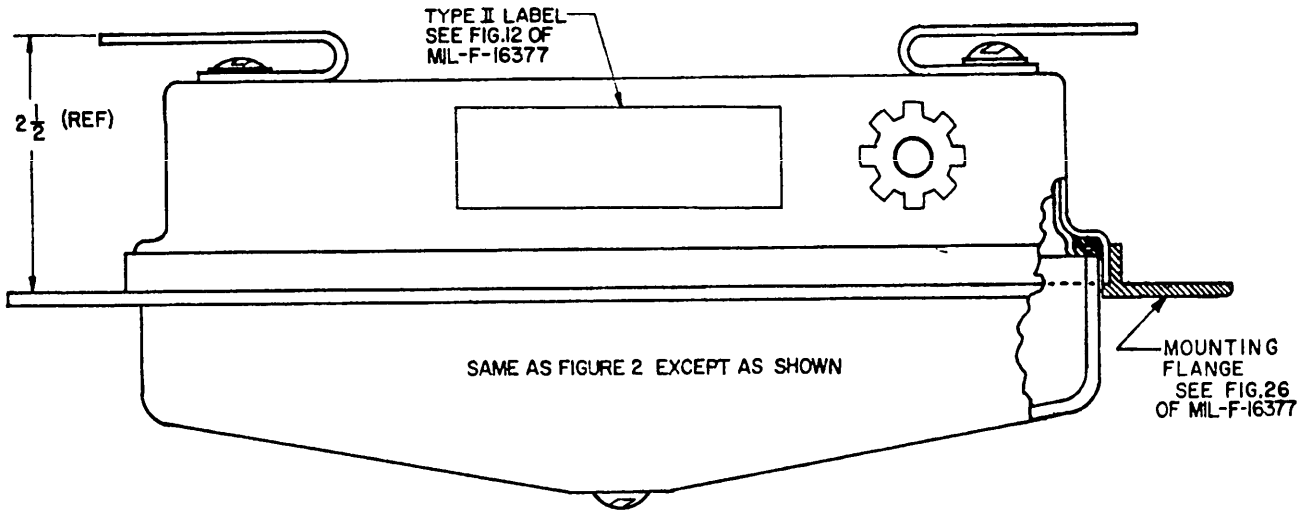
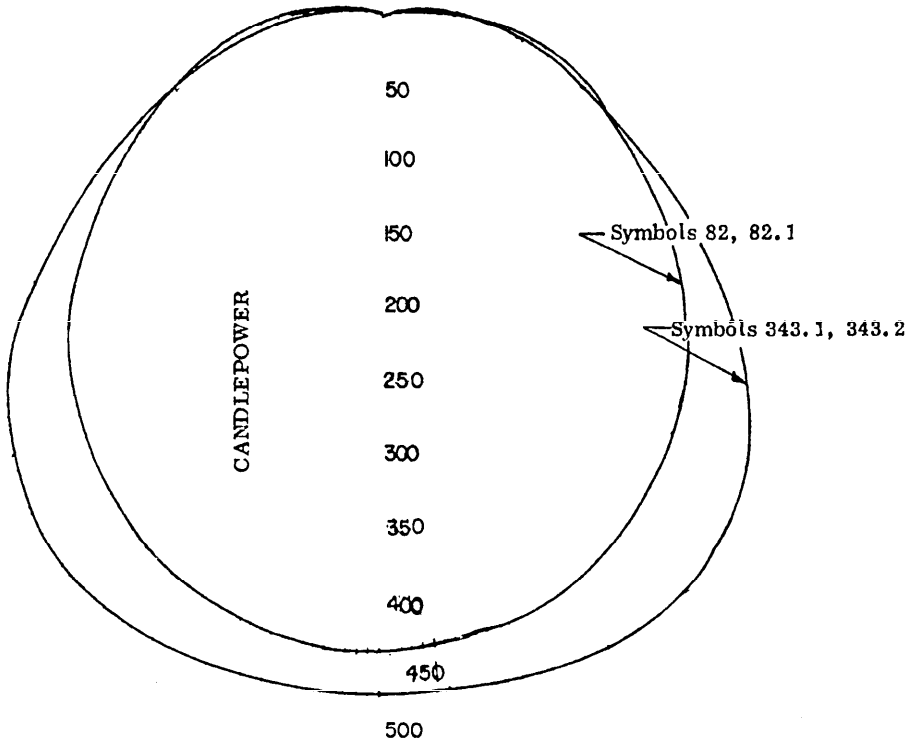


FIGURE 3. Typical section for Symbols 82.1 and 343.3.

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ANGLE	CANDLEPOWER SYMBOL NUMBER	
	82, 82.1	343.1, 343.2
0	440	460
5	430	460
10	425	460
15	420	460
20	410	455
25	390	450
30	370	440
35	350	415
40	330	390
45	300	360
50	270	320
55	235	280
60	210	225

ANGLE	CANDLEPOWER SYMBOL NUMBER	
	82, 82.1	343.1, 343.2
65	180	185
70	150	150
75	130	130
80	100	120
85	80	80
90	60	60
95	40	40
100	20	20
105	10	10
110	0	0
115	0	0
120	0	0
125	0	0

Photometric test data:

Distance: 10 feet.
Plane: At right angles to long axis of fixture through geometric center of lamp.
Voltage: 115 volts, 60 Hz.
Ambient temperature: 76°F.
Lamps: Use lamps furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp correction: Distribution curves based on 790 lumens per lamp.

FIGURE 4. Candlepower distribution curves.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 2

SYMBOL NUMBERS:
82 : Complete lighting fixture assembly, standard mounting, with three 15-watt cool white fluorescent lamps and white diffusing window assembly.
82.1: Complete lighting fixture assembly, flush mounting with three 15-watt cool white fluorescent lamps and white diffusing window assembly.
343.1: Complete lighting fixture assembly, standard mounting with three 15-watt cool white fluorescent lamps and clear and white prismatic window assembly.
343.2: Complete lighting fixture assembly, flush mounting, with three 15-watt cool white fluorescent lamps and clear and white prismatic window assembly.

DESIGN:
Electrical rating: 45 watts, 120 volts, 60 Hertz.
Dimension and configurations: See figures 1, 2 and 3 and figure 10 of MIL-F-16377.
Photometric data: see figure 4.
Lampholder spacing: see figure 21 of MIL-F-16377.
Wiring diagram: see figure 24 of MIL-F-16377.
Mounting dimensions:
Symbols 82 and 343.1 : see figure 25 of MIL-F-16377.
Symbols 82.1 and 343.2: see figure 26 of MIL-F-16377.
Window assembly:
Symbols 82 and 82.1 : Military Part Number M16377/66-001, figure 27 of MIL-F-16377.
Symbols 343.1 and 343.2:Military Part Number M16377/66-002, figure 27 of MIL-F-16377.
Enclosure: Watertight. See 3.5.11 of Mil-F-16377.
Cable: 10 feet of type SJO or SJT, in accordance with J-C-580 except AWG size 16, three conductor (white, black and red).
Weight:
Symbols 82 and 343.1 : 12 lbs 0 oz. maximum.
Symbols 82.1 and 343.2 : 13 lbs 10 oz. maximum.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figures 1 and 3:

(a) Two type I
(b) One type II for Symbols 82.1 and 343.2 only.
(c) One type IV. The information to be shown on the type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY." List all standard parts shown in table I and window assemblies of table IV by name and stock number only.

Standard Parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

QUALITY ASSURANCE:

First article inspection:
Complete fixtures: See table II
Window assemblies: See table III
Quality conformance: Same as the first article inspection

Inspection of product for delivery:
Examination: See MIL-F-16377
Examination for preparation of delivery: MIL-F-16377
Quality conformance tests: See MIL-F-16377

GENERAL INFORMATION:

Assignment of Military part numbers and general information is shown in table IV.

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TABLE I. Standard parts.

Name	Specification or Military part number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/13	6240-00-583-3668	Industry Number F15T12/CW
Lamp lock	M16377/40-002	6250-00-399-7040	
Lampholder	MIL-L-970/11	6250-00-284-0481	
Startersocket	MIL-L-970/13	6250-00-295-2738	
Starter	8W, Type III of W-S-755	6250-00-299-2884	Commercial Designation: FS2
Ballast	M16377/44-001	6250-00-569-9502	
Shockmount	M16377/41-009 or M16377/41-010	6210-00-973-1747 6210-00-973-1748	See 3.5.16 of MIL-F-16377
Red filter tube	M16377/42-004	6210-00-834-1621	Not furnished See 3.7.7 of MIL-F-16377

TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

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TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.4.1.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectivneess	3.5.11	4.8.14.4	Watertight
Light output	3.7.1	4.8.6.1	Candlepower

TABLE IV. Part number and general information.

Military Part Number M16377/66-	Item name	National Stock Number (NSN)	Remarks
82	Complete fixture, Symbol 82		New
82.1	Complete fixture, Symbol 82.1		New
343.1	Complete fixture, Symbol 343.1		New
343.2	Complete fixture, Symbol 343.2		New
001	Window assembly, white diffusing		New
002	Window assembly, clear and white prismatic		New

Preparing activity:
Navy - SH
(Project 6210-N570)

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